AI-Enabled Coconut GenBank Intelligence and Crowdsourced WCLWD Severity Assessment

IT4010 Research Project – 2026



R26-IT-019

The Problem



Users & Stakeholders

Coconut Farmers

CRI Authorities

Plantation Managers

Field Officers

System Administrators

Agricultural Researchers





Impact of the Problem on Users and Society

Impact on Farmers

- Late disease detection reduces crop yield and income.
- Dependence on experts slows treatment decisions.

Impact on Agricultural Planning & Disease Monitoring

- Lack of accurate GIS-based land analysis.
- Delayed detection and difficulty identifying diseases like WCLWD.

Impact on Research and Gene Bank Management

- Manual germplasm identification and fragmented records.
- Inefficient data management and slow research processes.

Impact on Community & Economy

- No unified platform for crowdsourced disease reports.
- Reduced coconut production impacts the national economy.

Existing Solutions and Their Limitations



Manual Gene Bank Management

Germplasm identification is done through manual observation, which is slow and difficult to scale.



Traditional Land Surveys

Plantation mapping relies on manual surveys without accurate GIS-based analysis.



Expert-Based Disease Monitoring

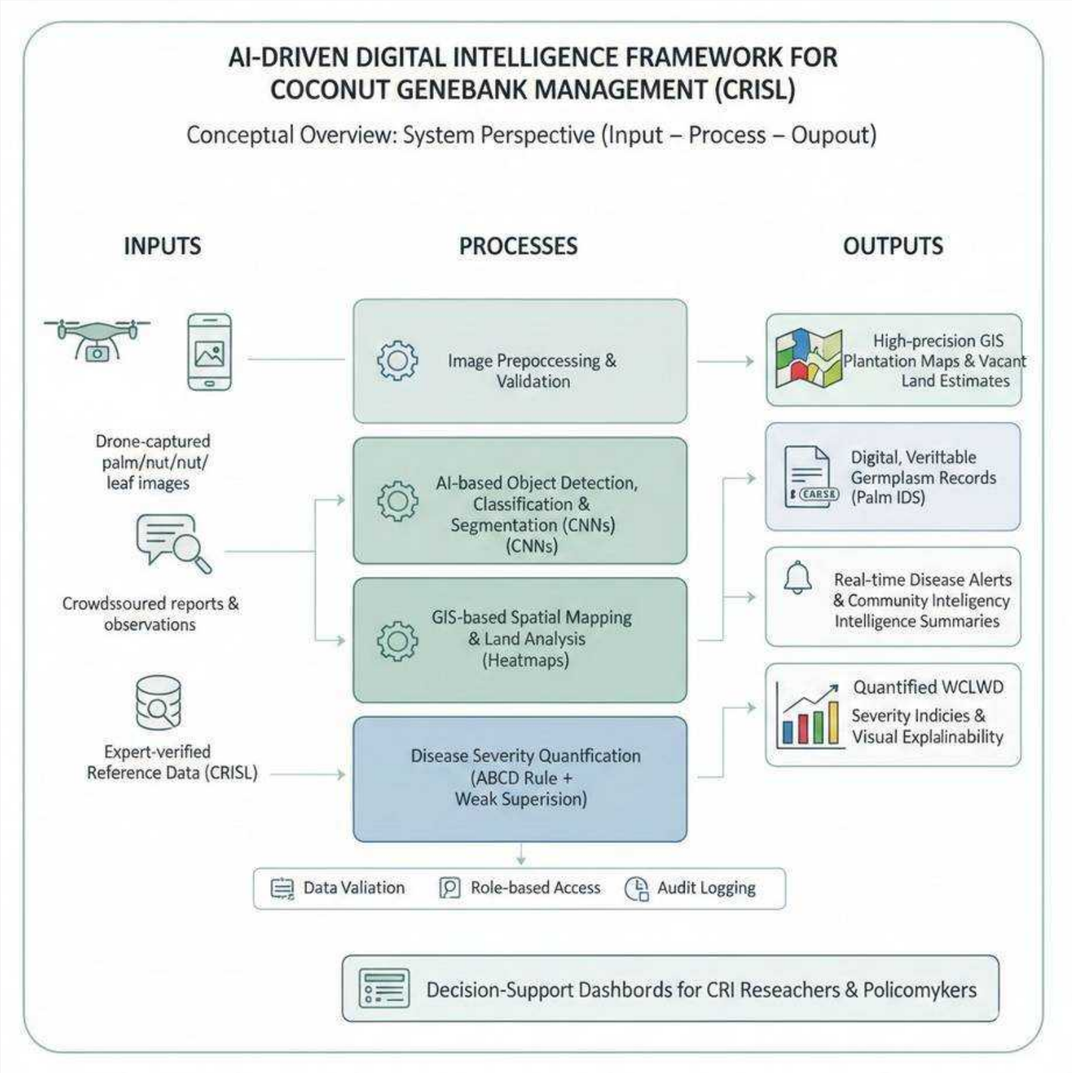
Disease detection depends on periodic expert inspections, causing delayed identification.



Isolated Agricultural Systems

Existing systems solve individual problems but lack an integrated AI-driven platform.

Proposed Solution



The proposed solution is a **“AI-Enabled Coconut GeneBank Intelligence and Crowdsourced WCLWD Severity Assessment”**



IT22904164 | Wijerathna W.M.H.N.
GIS-Based Land Utilization Analysis System for Coconut Genebanks

Contribution to Overall Solution

This component provides the geospatial analysis and visualization module of the overall system. It analyzes spatial data collected from coconut genebank locations to identify utilized plantation areas and available land spaces.

Novelty

- Introduces GIS-based spatial analysis for coconut genebank land utilization
- Uses heatmap visualization to represent land usage intensity across plantation areas
- Integrates spatial datasets such as plantation boundaries, tree coordinates, and yard space information
- Provides visual decision-support tools for researchers and plantation planners

⚠ Problem Definition

- **Land utilization in coconut genebanks is currently assessed using manual records and tabular data.**
- **Lack of spatial visualization makes it difficult to interpret plantation distribution.**
- **Identifying underutilized land and available planting areas is time-consuming.**

Study	GIS Approach	Land Utilization Analysis	Heatmap Visualization	Drone/ Satellite Data
Sawant et al. (2023)	✔	✘	✘	✔
Rahim et al. (2024)	✔	✘	✘	✔
Chen et al. (2021)	✔	✘	✘	✔
Proposed System	✔	✔	✔	✔

Objective

Develop a GIS-based system that generates heatmaps to analyze land utilization in coconut genebanks.

Sub-Objectives

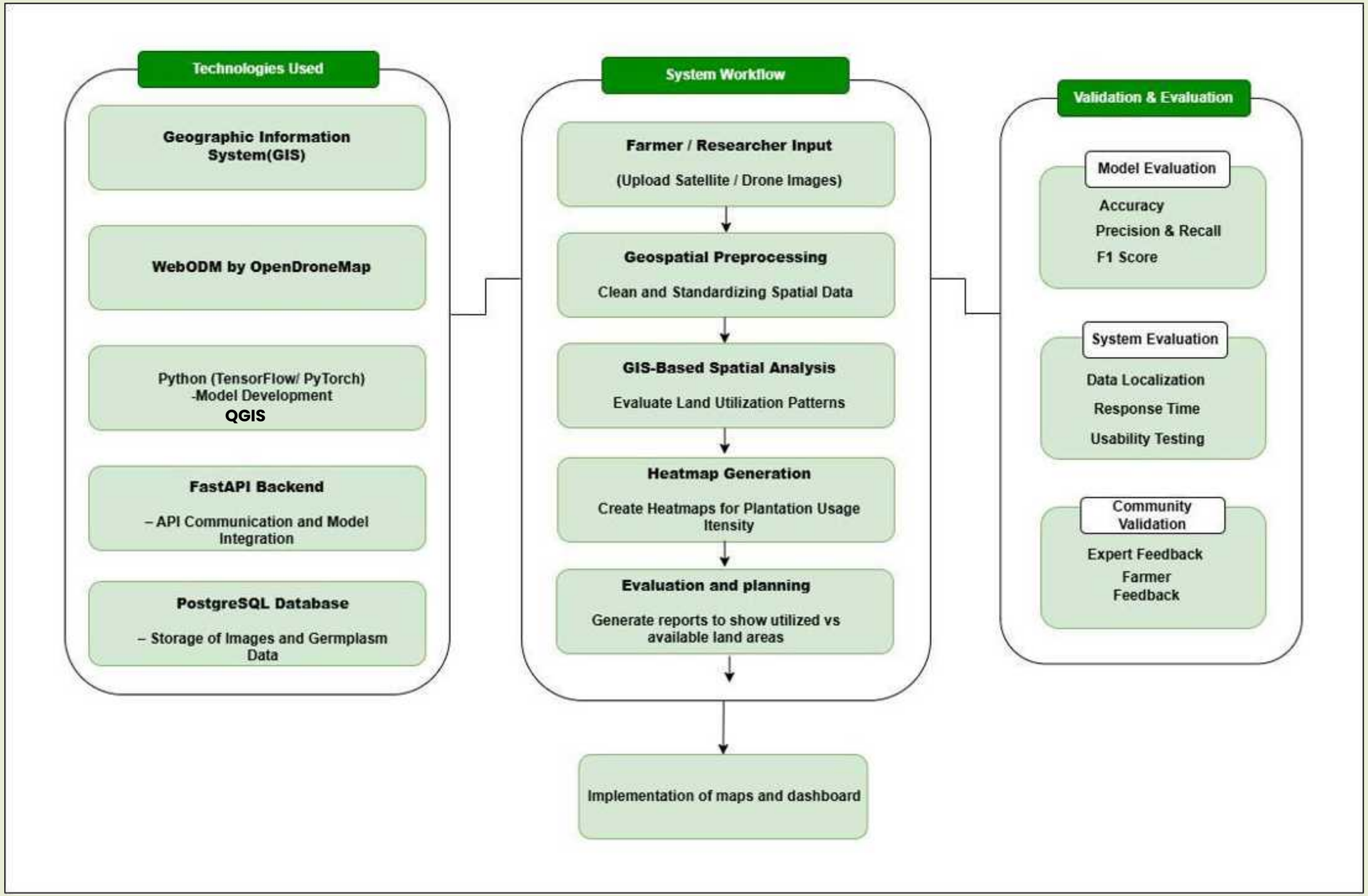
- Collect satellite and drone-based spatial
- Apply GIS spatial analysis.
- Generate heatmap visualizations
- Identify underutilized areas suitable
- Provide a visual dashboard

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GIS-Based Land Utilization Analysis System for Coconut Genebanks

Research Methodology



Data Availability

- Spatial data obtained from satellite imagery and drone images
- Includes plantation boundaries, tree locations, and land areas
- Images converted into GIS-compatible spatial datasets

Data Security

- Data stored in secure research storage systems
- Access restricted to authorized researchers

Ethical Clearance

- Data used only for academic research purposes
- No personal or sensitive information collected

GIS-Based Land Utilization Analysis System for Coconut Genebanks

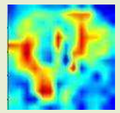
SYSTEM OVERVIEW

Collect satellite and drone images of coconut genebank



Perform GIS-Based spatial analysis to evaluate land utilization patterns

Generate heatmaps to visualize utilized areas and available plantation spaces



Display analysis results through the web-based GIS dashboard



Dependencies and Common Components



Satellite and drone imagery datasets for spatial analysis



GIS tools and libraries for geospatial processing



FastAPI backend for system communication.



Backend APIs for data processing and postgreSQL database for storing spatial data.



Python (TensorFlow / PyTorch) for model development.

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IT22236500 | Randeniya A.A.S.L.B.R.P.W.R.C

A Secure Image Classification system for coconut Germplasm Identification

Contribution to Overall Solution

This component provides the AI classification engine that automatically identifies coconut germplasm types from images and generates digital gene bank records.

Novelty

- First AI system targeting coconut germplasm identification
- Combines CNN + Few-Shot Learning + Grad-CAM explainability
- Provides a secure digital platform for gene bank data



⚠ Problem Definition

- Germplasm identification relies on manual observation of morphological traits.
- The process requires expert knowledge and is difficult for farmers.
- Manual identification is time-consuming and hard to scale.
- Existing AI research focuses on disease detection, not germplasm identification.

Study	Image Classification	Germplasm Identification	Explainable AI	Farmer-Accessible System
Mohanty et al. [1]	✓	✗	✗	✗
Lee et al.[2]	✓	✗	✗	✗
Pourreza et al., [3]	✓	✓ (Wheat)	✗	✗
Our Solution	✓	✓	✓	✓

Objective

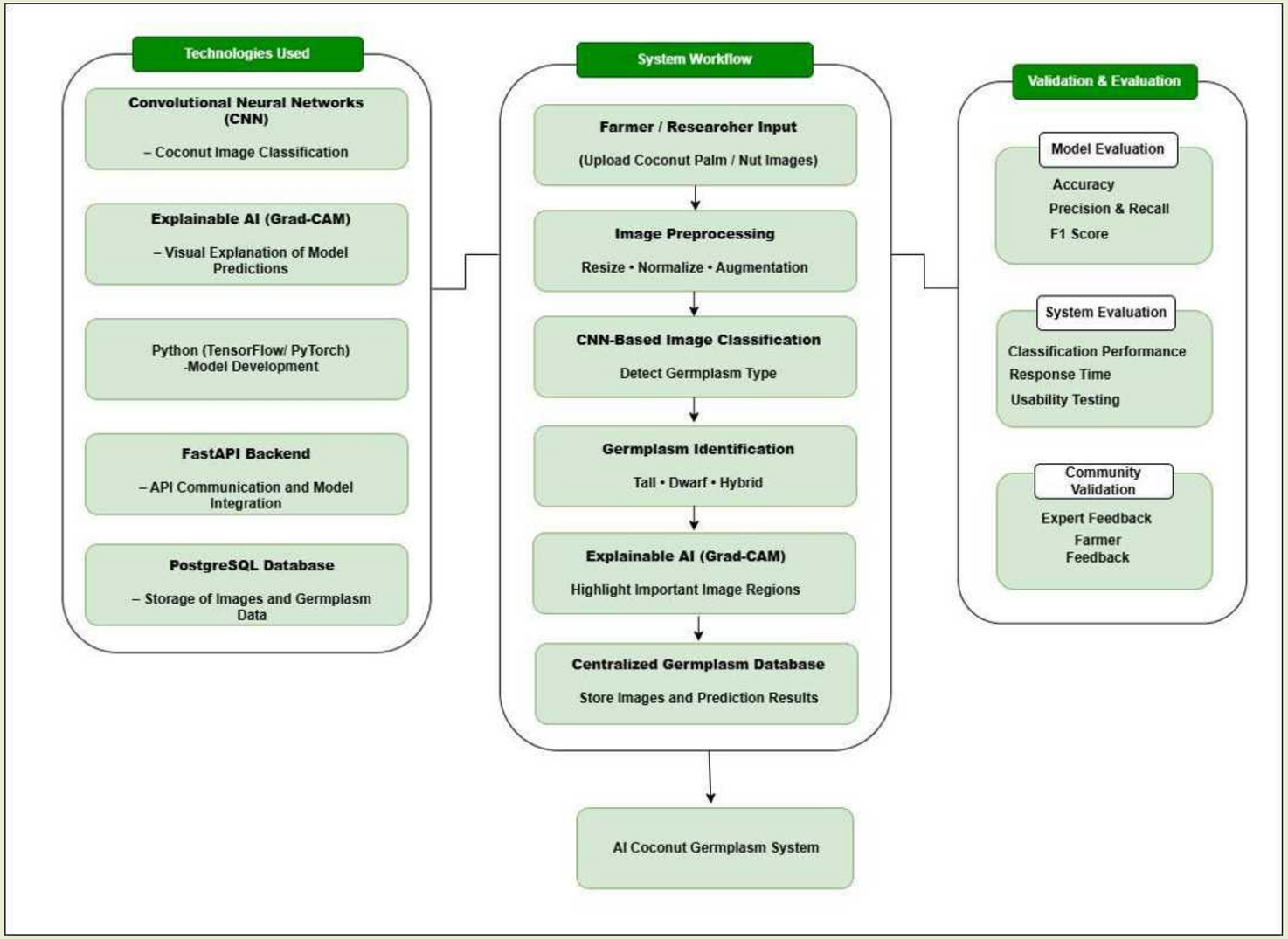
Develop an AI-based system to identify coconut germplasm from images.

Sub-Objectives

- Build coconut germplasm image dataset
- Develop CNN classification model
- Apply few-shot learning
- Implement Grad-CAM explainability
- Develop web-based identification platform

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Research Methodology



A Secure Image Classification system for coconut Germplasm Identification

Data Availability

- Coconut palm and nut images collected from CRI gene banks and plantations
- Dataset contains images representing Tall, Dwarf, and Hybrid germplasm types

Data Security

- Role-based access control for researchers and users
- Secure API communication for data transfer
- Database protection and encrypted storage

Ethical Clearance

- Permission obtained for using agricultural research data
- No personal or sensitive data collected from users
- Dataset used strictly for research and agricultural development

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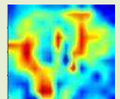
A Secure Image Classification system for coconut Germplasm Identification

SYSTEM OVERVIEW

Upload coconut palm / nut images through the system.



Images are processed and classified using a CNN-based AI model.



Grad-CAM explainable AI highlights important regions used for prediction.



stored and displayed through the web platform.

Dependencies and Common Components



Image dataset from coconut plantations / gene banks.



Python (TensorFlow / PyTorch) for model development.



FastAPI backend for system communication.



PostgreSQL database for storing germplasm data.

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IT22549204 | Harischandra R.D.I.D

Crowdsourced Community Intelligence Platform for Coconut Caterpillar Infestation Monitoring

Contribution to Overall Solution

Develops a crowdsourced community intelligence platform where farmers report coconut caterpillar infestations using images and text. The system integrates CNN-based image detection, lightweight NLP analysis, community knowledge sharing, and geographic visualization for early pest monitoring

Novelty

- Crowdsourced community intelligence platform for coconut caterpillar monitoring
- CNN-based image classification for detecting caterpillar damage
- Lightweight NLP analysis of farmer pest reports and queries
- Community Q&A with confidence-based answer ranking
- Interactive dashboards and geographic heatmaps for pest monitoring



⚠ Problem Definition

- **Coconut caterpillar infestation significantly damages coconut leaves and reduces plantation productivity.**
- **Pest monitoring mainly relies on manual field inspections and expert observation.**
- **Farmers have limited access to timely pest identification and expert guidance.**
- **Existing systems rarely combine AI detection, crowdsourced reports, knoweledge sharing and geographic monitoring.**

Study	AI-Based Pest Detection	Crowdsourced Field Data	NLP-Based Farmer Report Analysis	Geographic Pest Monitoring
PlantVillage CNN Disease Detection System (2016)	✓	✗	✗	✓
Crowdsourcing-Based Pest Surveillance Platform (2021)	✗	✓	✗	✓
Deep Learning Coconut Disease & Caterpillar Detection System (2025)	✗	✗	✗	✓
Proposed Crowdsourced Coconut Caterpillar Monitoring Platform	✓	✓	✓	✓

Objective

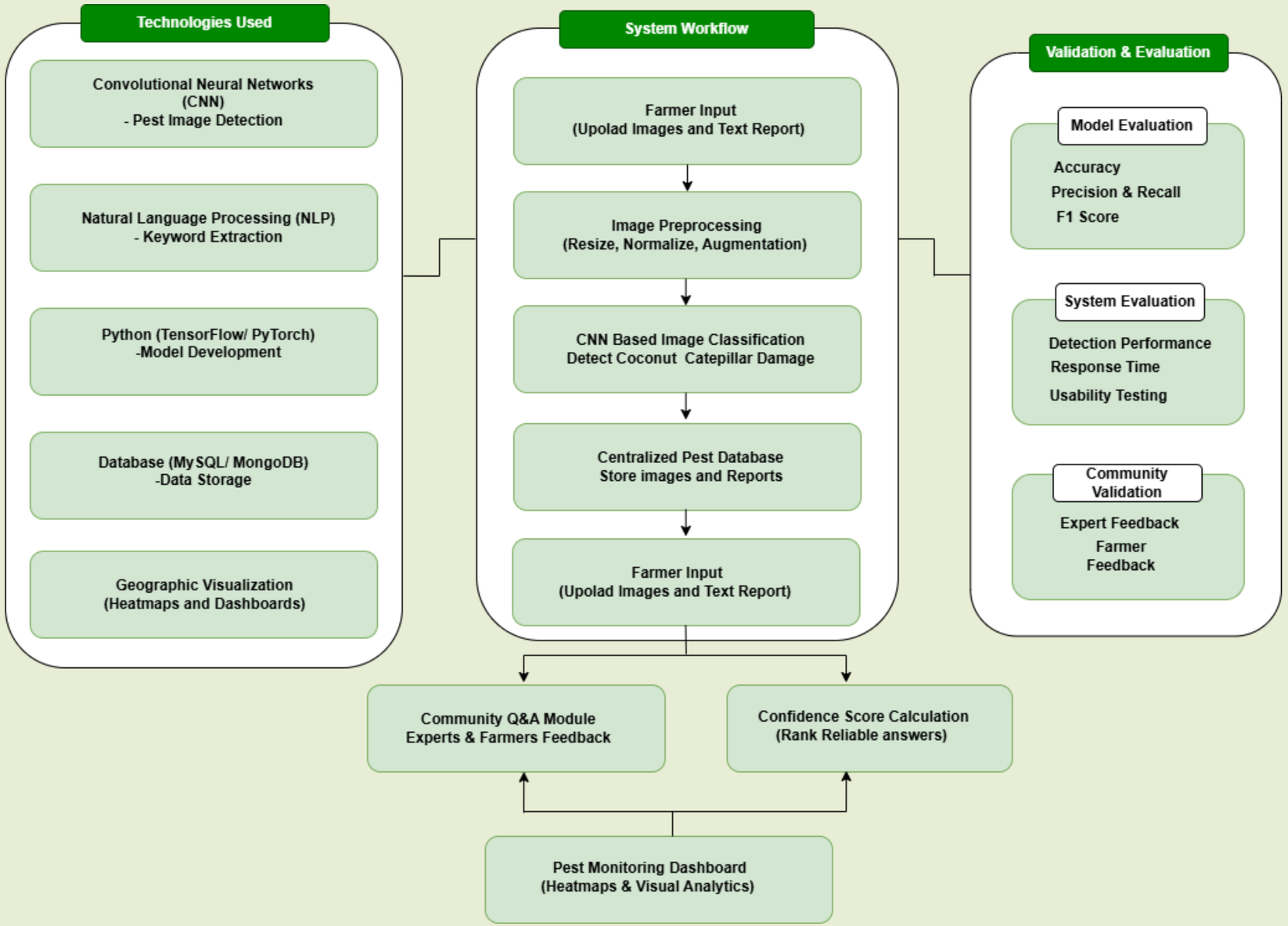
Develop a crowdsourced community intelligence platform for coconut caterpillar detection, monitoring, and knowledge sharing.

Sub-Objectives

- Develop a crowdsourced web platform
- Implement CNN-based image analysis
- Apply lightweight NLP techniques
- Visualize pest data through dashboards and geographic heatmaps
- Implement a knowledge sharing platform

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Research Methodology



Data Availability

- Coconut pest image dataset obtained from Kaggle
- Dataset originally collected from coconut plantations in Sri Lanka
- Includes images showing coconut caterpillar infestation symptoms
- Images captured under different environmental conditions

Data Security

- Data stored in a secure research database
- Access limited to authorized research team members
- Uploaded farmer reports, images are protected from unauthorized access

Ethical Clearance

- No personal or sensitive farmer information collected
- Images used only for research and pest monitoring purposes
- Study follows academic research ethics and data privacy guidelines

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Crowdsourced Community Intelligence Platform for Coconut Caterpillar Infestation Monitoring

SYSTEM OVERVIEW



Farmers upload images and pest Reports



CNN Image Analysis(Pest Detection) CNN Model Analyzes images to detect suspected coconut caterpillar images



NLP Report Analysis(NLP analyzes textual reports to extract pest information)



Community Q&A Platform(enables knoweledge Sharing)



Pest Monitoring Dashboard + Visualization heatmaps

Dependencies and Common Components



Kaggle Coconut Pest Dataset for Model Training Python (TensorFlow/ PyTorch)



Spacy/NLTK



Web Platform (React/ FastAPI)for User Interaction



MYSQL database for storing reports and images



Leaflet.js /PowerBI for heatmap and dashboard visualization

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IT22088314 | Malshani W.A.M.S

Weligama Coconut Leaf Wilt Disease Severity Assessment Using ABCD Rule

Contribution to Overall Solution

This component develops an AI -based severity assessment module that automatically detects diseased regions in coconut leaf images and estimates the severity of WCLWD infection using image weakly supervised learning.

Novelty

- Introduces an automated severity estimation framework for WCLWD using leaf images
- Combines Vision Transformer + Grad CAM + Weakly Supervised Segmentation
- Utilizes the ABCD rule to extract lesion morphological features
- Provides quantitative disease severity scoring for plantation monitoring



⚠ Problem Definition

- **WCLWD causes severe leaf damage such as flaccidity, yellowing, and browning, leading to reduced coconut yield and plantation productivity**
- **Traditional detection methods (PCR, ELISA) require laboratory facilities and are not suitable for real-time field monitoring.**
- **Existing AI methods mainly classify disease presence and require manual pixel-level annotations, making severity assessment difficult and costly.**

Research Gap

Study	Disease Severity Estimation	Scalability	Automated Annotation
Nested PCR and qPCR Research	✗	✗	✗
Early Diagnosis of WCLWD & Caterpillar Using Deep Learning	✗	✗	✗
Automatic Pixel-Level Annotation for Severity Estimation	✗	✗	✓
Proposed System	✓	✓	✓

Quantitative Severity Estimation | Scalable & Field-Ready | Automated & Efficient

Objective

Develop an AI-based system to detect WCLWD and estimate disease severity from coconut leaf images.

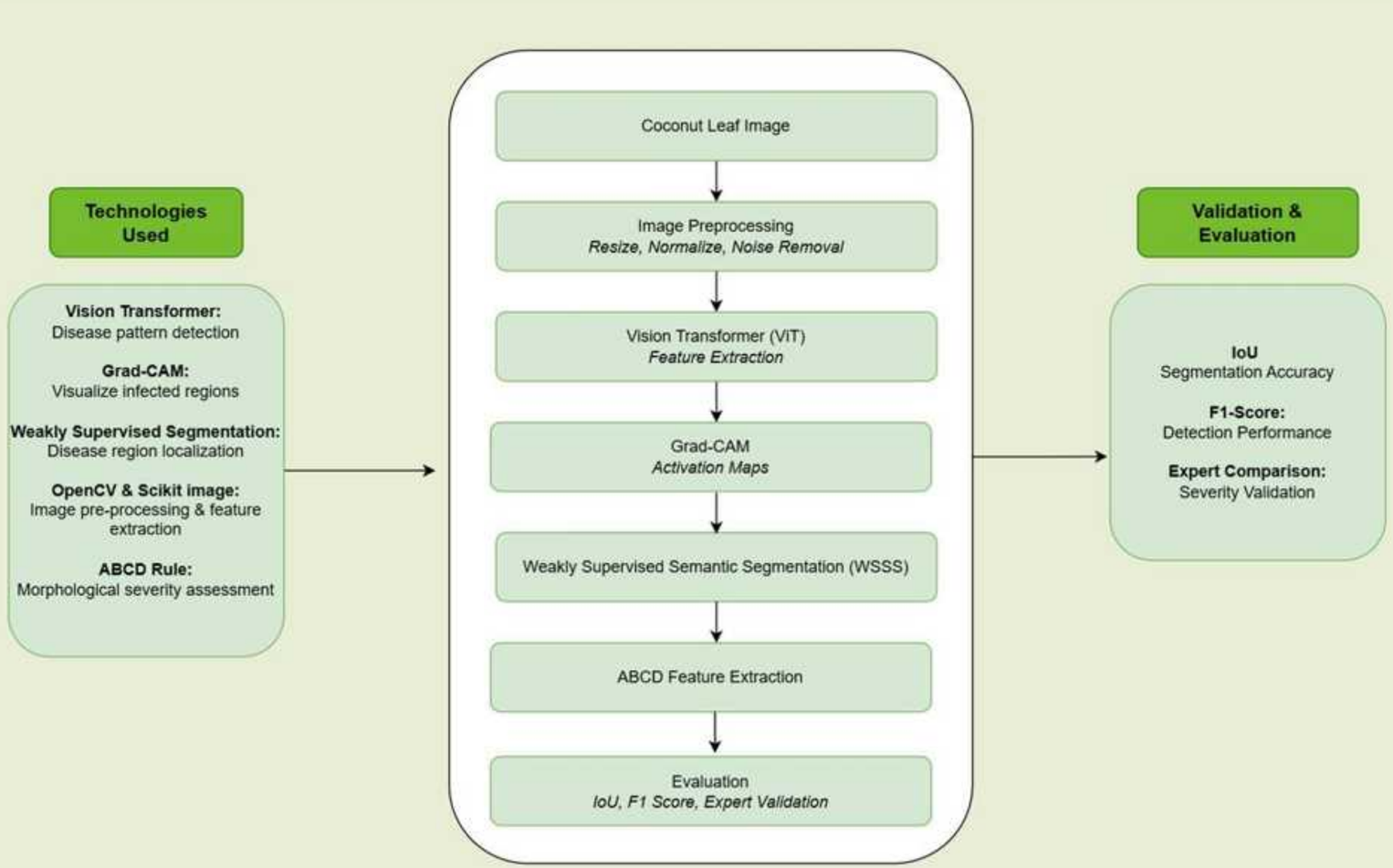
Sub-Objectives

- Leaf image preprocessing
- ViT disease detection
- WSSS lesion segmentation
- ABCD feature extraction
- Disease Severity Estimation

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IT22088314 | MALSHANI W.A.M.S | R26-IT-019

Research Methodology



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Data Availability

- Existing coconut leaf image dataset obtained from Kaggle
- The dataset contains images showing WCLWD symptoms such as:
  - Flaccidity
  - Yellowing
  - Leaflet tip browning
- Images captured using smartphones and DSLR cameras

Data Security

- Data stored in secure research storage
- Access restricted to research team members only

Ethical Clearance

- No personal or sensitive information included
- Research complies with academic ethical guidelines

Weligama Coconut Leaf Wilt Disease Severity Assessment Using ABCD Rule

SYSTEM OVERVIEW

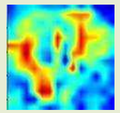
Upload coconut leaf images.
Images of coconut leaves are uploaded through the web-based system



Images Processing
Images are resized, normalized and noise is removed using image processing techniques



Vision Transformer Detection
The ViT model extracts disease-related visual features from leaf images



Grad-CAM Activation Mapping
Grad-CAM highlights the important regions contributing to disease detection



Weakly Supervised Segmentation
Diseased regions are localized without manual pixel-level annotations

Dependencies and Common Components



Coconut Leaf Image Dataset (Kaggle/ Research Dataset)



Python Deep Learning Framework (PyTorch)
Used to implement ViT and segmentation models



Image Processing Libraries
OpenCV and Scikit-image used for preprocessing and feature extraction



Web-based Monitoring Interface
Displays severity percentage and severity category for WCLWD



Severity Scoring System
ABCD rule-based Disease Severity Index (DSI) for quantitative disease assessment

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AI-Enabled Coconut GenBank Intelligence and Crowdsourced WCLWD Severity Assessment

Commercialization Potential

- Scalable AI platform for coconut disease monitoring and genebank management
- Can be adopted by CRI, agricultural institutes, and plantations
- Expandable to other crops and smart agriculture systems

Target Market & Customers

- Coconut Research Institute (CRI)
- Plantation owners & estate managers
- Farmers

Value Proposition

- GIS plantation mapping & decision support
- AI-based germplasm identification
- Real-time crowdsourced disease monitoring
- Automated WCLWD severity estimation

Cost & Revenue Model

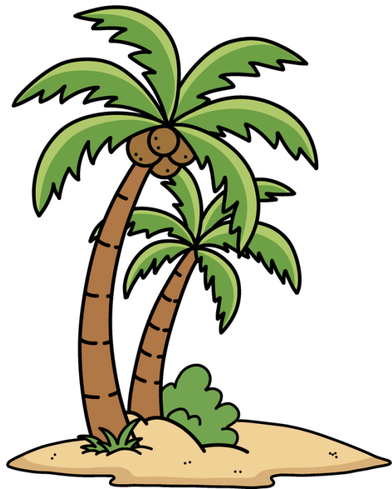
- **Investment:** AI Development, Cloud Infrastructure, Data collection
- **Recovery:** Institutional licensing, Subscriptions, Government Agricultural Programs



AI-Enabled Coconut GenBank Intelligence and Crowdsourced WCLWD Severity Assessment

GANTT CHART

TASK	OCT	NOV	DEC	JAN	FEB	MAR	MAY	JUN	JUL	AUG	SEP	OCV	DEC
1. Group Registration			20th October 2025										
2. Topic Selection													
3. TAF Documentation													
4. TAF Submission				05th January 2026									
5. Proposal Report					15th March 2026								
6. Proposal Present					16th March 2026								
7. Implementation													
8. Progress Presentation I							4th - 6th May 2026						
9. Research Paper													
10. Testing & Finalize the Project										12th - October 2026			
11. Progress Presentation II										12th - 14th October 2026			
12. Web Assessment											11th October 2026		
13. Final Presentation & Viva													1st December 2026
14. Final Report													



AI-Enabled Coconut GenBank Intelligence and Crowdsourced WCLWD Severity Assessment

Work Breakdown Structure (WBS)

AI-Enabled Coconut GenBank Intelligence System

GIS-Driven Land Utilization Analysis

- **1.1** Requirement Analysis
- **1.2** Satelitte / Drone Image Collection
- **1.3** Image Preprocessing
- **1.4** Coconut Tree Detection
- **1.5** GIS Mapping & Spatial Analysis
- **1.6** Plantation Density Heatmap Generation
- **1.7** Vacant Land Identification

AI-Based Germplasm Identincation System

- **2.1** System Requirement Analysis
- **2.2** Coconut Image Dataset Collection
- **2.3** Image Preprocessing & Augmentation
- **2.4** CNN Model Development
- **2.5** Few-Shot Learning Implementation
- **2.6** Germplasm Classification
- **2.7** Grad-CAM Explainable AI Visualization
- **2.8** Digital Gene Bank Record Generation

Crowdsourcing Community Intelligence Platform

- **3.1** System Architecture Design
- **3.2** Community Data Collection Platform
- **3.3** Image & Text Report Processing
- **3.4** Disease Detection Model
- **3.5** Knowledge Mapping & Data Analysis
- **3.6** Community Dashboard & Insights
- **3.7** Real-Time Monitoring Interface

WCLWD Disease Severity Assessment System

- **4.1** Dataset Preparation
- **4.2** Image Segmentation Model
- **4.3** Weakly Supervised Learning Implementation
- **4.4** Diseased Region Detection
- **4.5** ABCD Severity Rule Application
- **4.6** Severity Index Calculation
- **4.7** Visualization & Decision Support Dashboard



THANK YOU!

